DATA STRUCTURES & ALGORITHMS DEFENSE[1]

OPTION 1:

Answer any TWO uestions. Each question carries equal marks.

1. Write a program to find the smallest odd number in an array.
2. Write a program to delete an element at a specific position in an array.
3. Explain and implement a sorting algorithm of your choice (e.g., Bubble Sort, Selection Sort).
4. Given an array of integers (possibly with duplicates), write a program to find the second largest element.
5. Write a menu-driven program with exactly three operational options plus an Exit option. After completing any operation, redisplay the menu and continue until the user selects Exit. The program must validate input: reject non-integer entries and out-of-range choices by displaying a clear error message and re-prompting without terminating.
6. Write a program to check whether an array is sorted. If it is, state whether it is in ascending or descending order.
7. Given a sorted list and a new value, write a program to insert the value at the correct position.
8. Implement a search() function to return the index of the last occurrence of an element in an array of strings.
9. Define a singly or doubly linked list and write programs to insert a new node at (a) the beginning, and (b) the end of the list without using Java's built-in LinkedList class.
10. Implement a stack using an array with the operations push() and pop() without using Java's built-in Stack class.
11. Implement a queue using an array with the operations enqueue() and dequeue() without using Java's built-in Queue interface.
12. Write a program to reverse an array in place (without using another array).
13. Write a program to count the frequency of each element in an integer array.
14. Write a program to identify the missing number in a sequence of integers ranging from 1 to n (where exactly one number in the range is absent)

OPTION 2:

Answer all the following questions.

1. (2 marks) Create a 'Student' class with attributes such as id, name, course, and grade. Then, create a list of N students.
2. (4 marks) Implement the following functions:
   - Find students enrolled in a specific course.
   - Count the number of students who achieved each grade (e.g., A, B, C, etc.).
3. (4 marks) Implement the following functions:
   - Find the student(s) with the highest grade.
   - List all students whose grade is below the average grade of the class.

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[1] Hanoi, Aug. 2025. Contact: quandh13@fe.edu.vn